RIGHT FRACTIONS FORM A CATEGORY VIA THE OPPOSITE CATEGORY DICTIONARY

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4811-4817, following the statement at 4798-4809.

Editorial note. The proof is a complete opposite-category dictionary reduction to the cited lemma (carded separately with a fully executed proof); the three claim components translate mechanically under arrow reversal, recorded in the card's obligations.

Allowed standard prerequisites (editorial).

- Left fractions form a category. lemma-left-localization (carded separately with a complete proof), cited at proof lines 4812-4813. Supplies the three-part statement (equivalence relation, well-defined composition, associativity and identities) for left multiplicative systems, applied to the opposite category.
- Right multiplicative systems become left systems in the opposite category. the right multiplicative system axioms of the statement, used at proof lines 4814-4816. The axioms are self-dual under arrow reversal, so S is a left multiplicative system in C^opp; the right fraction constructions translate to left fraction constructions in C^opp.

Source reference. Original lemma 04VD: sources/stacks-categories/pinned-original-source.tex, source lines 4219–4219. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a category and let S be a right multiplicative system.*

(1) *The relation on pairs defined above is an equivalence relation.*
(2) *The composition rule given above is well defined on equivalence classes.*
(3) *Composition is associative (and the identity morphisms satisfy the identity axioms), and hence $S^{-1}\mathcal{C}$ is a category.*

Proof. This lemma is dual to Lemma 04VD. It follows formally from that lemma by replacing $\mathcal{C}$ by its opposite category in which S is a left multiplicative system. □

Date: stacks-project:categories.tex:local-snapshot-20261012; GNU FDL 1.2 or later.